Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		[The sources have] constant phase {difference / relationship} Don't accept in phase	1			1		
	(b)	(i)	The difference in the distance that the two waves travel [to reach the same point] (1) Accept S_1B - S_2B At <u>first</u> maxima wave from S_1 to B has travelled λ metres further (1)	1	1		2		
		(ii)	**Any 2 ×(1) from:** 0.3 [m] / 0.9 [m] / 1.5 [m] / ….. Award 1 mark for $\frac{\lambda}{2}$ **and** $\frac{3\lambda}{2}$ or equivalent stated with no values given		2		2		
	(c)	(i)	If waves [from two sources] {overlap / cross / interfere} then the total displacement [at any one point] (1) is the [vector] sum of their individual displacements [at that point] (1)	2			2		
		(ii)	@ A and B: Resultant amplitude / total max displacement is less or algebraic / numerical argument i.e. $A_{S1} + A_{S2\text{ fault}} < A_{S1} + A_{S2\text{ orig}}$ (1) @ C: Resultant amplitude / total max displacement is greater or algebraic / numerical argument i.e. $A_{S1} - A_{S2\text{ fault}} > A_{S1} - A_{S2\text{ orig}}$ (1) Accept: Total destructive interference becomes partial destructive interference as waves no longer cancel out. {Waves continue to arrive in phase ***/*** constructive interference} at A and B **OR** {waves continue to arrive in anti-phase / destructive interference} at C (1)		3		3		
			Question 4 total	**4**	**6**	**0**	**10**	**0**	**0**